# Predict Before You Modify: Learning Improvement Dynamics for Self-Evolving Agents

Zixuan Xie, Chang Xu, Yuhua Jiang, Li Zhao · Microsoft Research · UVA · Tsinghua · arXiv 2610.13991v2

HarnessRSI 72 ★★★

evidence: improver

Paper

Code (anon.)

## 1 What changes, what is kept

**HyperAgents + one new editable part**

| | | | |
|---|---|---|---|
| **Task agent** | solves the target tasks | **Meta agent** | edits task agent, itself, IDM |
| **IDM (new)** | frozen LLM + editable code/prompts | **Archive** | evaluated agents = parents |

**1 Meta agent drafts a diff**
parent drawn by the fixed HyperAgents rule; the draft is a code change to the parent

**2 IDM forecasts the draft**
per-task outcome probabilities (e.g. resolved / incomplete / empty patch) → predicted score

**3 Revise once? (fixed rule)**
bias-corrected predicted score < parent's score: one extra revision with the forecast; else keep

**4 Always evaluate the child**
final diff forecast again; (forecast, outcome) record stored; child joins the archive

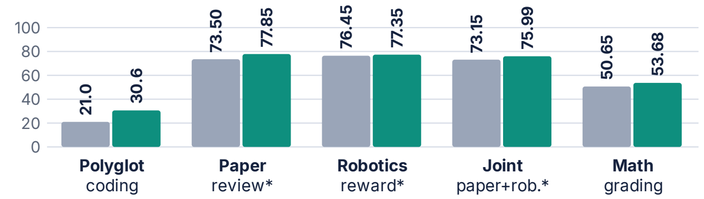
**5 Every K = 5 gens: edit the IDM**
candidate kept only if loss is lower on the next K records and CMSE does not rise

40×

**5 runs × 40 generations per setting; medians**
LLM weights never change; kept artifacts are code and prompts

The child is always evaluated: forecasts steer a revision, never skip an evaluation. Each setting: same task/meta LLMs, splits and parent rule as the baseline.

## 2 Headline: higher median AUBC in 4 of 4




■ HyperAgents (baseline) ■ **Selective IDM (ours)** · 40-gen AUBC ×100 (Polyglot 0.210 / 0.306 × 100), median of 5 runs. *Same joint runs. 95% intervals are in the paper (not drawn); most pairs overlap.

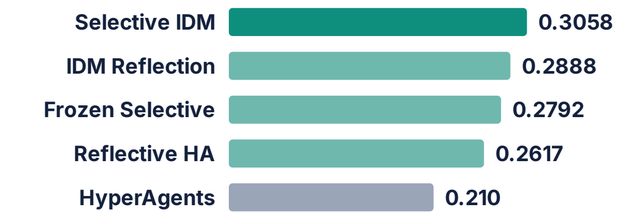
| Base → Selective | Coding (166) | Paper | Robotics | Joint | Math (100) |
|---|---|---|---|---|---|
| matched tok. | +10.9% | 72.00→74.95 | 75.08→75.79 | 71.79→72.46 | 51.05→53.56 |
| held-out | 5.42→13.25 pp | none | none | none | 47→51% |

## 4 Where the gain is absent or thin

- **Robotics:** 76.45 vs 77.35, 75.08 vs 75.79 (matched): < 1 point
- **All six** Selective-IDM-vs-ablation intervals (Table 7) include 0
- **Plain extra revision (no IDM)** recovers ≈54%‡ of the coding AUBC gap (0.2617 vs 0.210→0.306); full evaluation 35.0% vs 45.0% (Selective), 5.0% (HyperAgents)
- **Cost:** 170.22M vs 49.45M tokens (coding), +23%‡ vs Reflective HA; 140.61M vs 101.67M (joint)
- No held-out for paper / robotics; always-revise control only on Polyglot; one donor IDM

## 3 Controls that decide if the gain is real

### Ablation · Polyglot AUBC, 40 gen (Fig. 7, 5 runs)




### Same, as differences ×100 [95% CI] (Table 7)

| Selective IDM minus … | 40 gen | matched tok. |
|---|---|---|
| Reflective HA | 4.42 [−6.54, 9.25] | 1.21 [−6.32, 4.79] |
| Frozen Selective | 2.67 [−3.29, 7.62] | 0.92 [−4.73, 2.64] |
| IDM Reflection | 1.71 [−14.17, 5.46] | 1.13 [−9.78, 2.85] |

Cleanest IDM-feedback test, IDM Reflection vs Reflective HA: 2.7‡ points, no interval given. Budget 145.79M tokens. Candidate IDM lower Brier in 19 of 28 comparisons, the same block that gates adoption.

### Transfer · one IDM evolved on paper+robotics

| Math grading | HyperAgents | Selective | Transferred |
|---|---|---|---|
| AUBC ×100 | 50.65 | 53.68 | **55.52** |
| matched tokens | 51.05 | 53.56 | **54.81** |
| held-out acc. (100) | 47% | 51% | **55%** |

4 of 5 transfer runs exceed the Selective median. Donor-IDM evolution cost is not in these token budgets.

## Takeaway

Forecasting per-task outcomes before evaluation, with the forecaster itself rewritten over time, lifts median best-so-far scores in all four domains by a few points. The ablation order agrees, but its intervals include zero, held-out sets exist only for coding and math, and runs use about 1.4–3.4×‡ the tokens.

## Our read · HarnessRSI

**Score 72 · ★★★ · evidence level: improver**

**Reports:** held-out (coding, math), transfer, 5 runs + CIs, matched-token AUBC, cost, ablations.

**Does not report:** always-revise control beyond Polyglot, contamination audit.